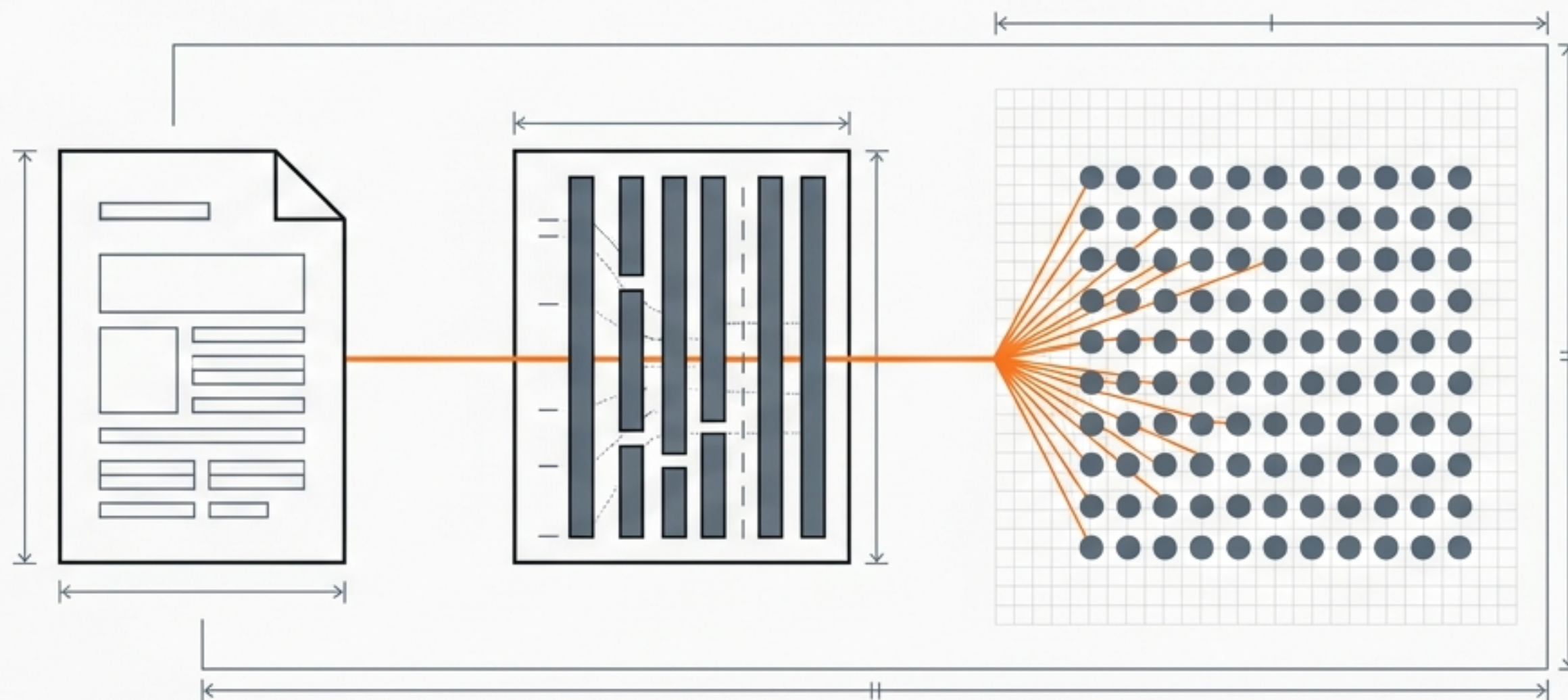


Building the Ingestion Pipeline

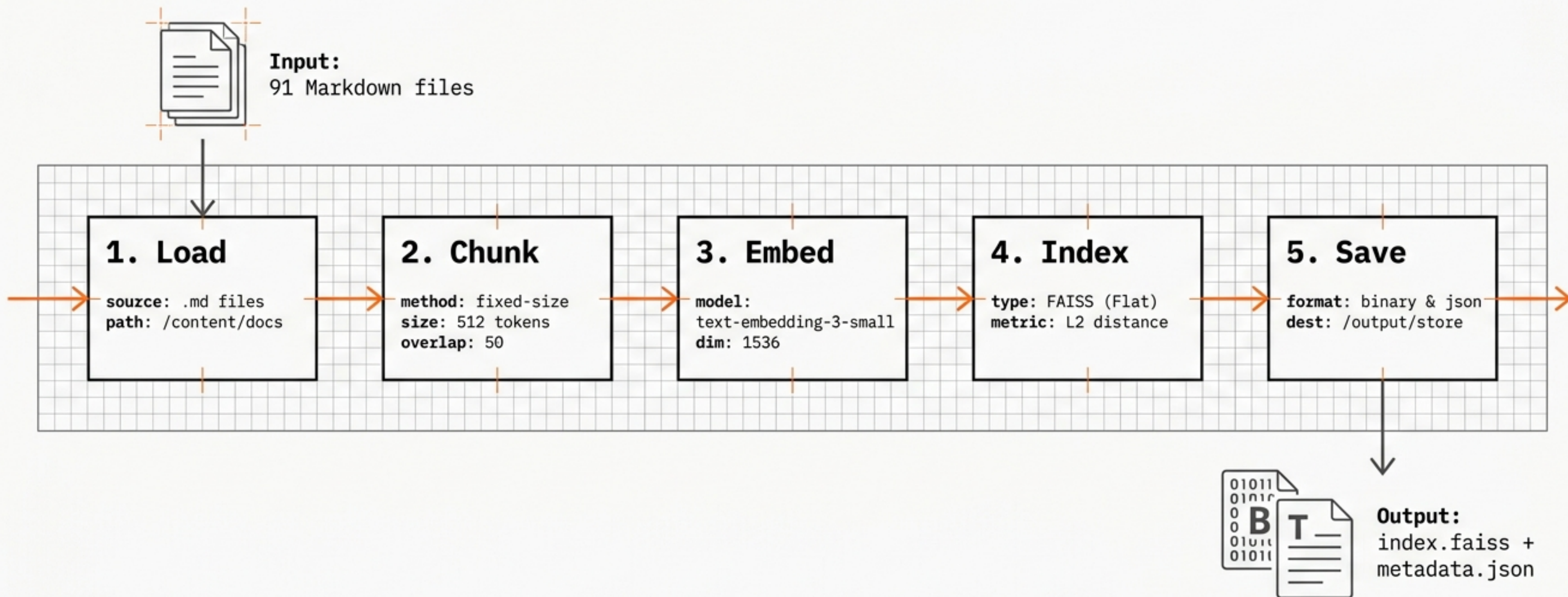
Phase 1 of the Holberton RAG System

Transforming 91 static documents into a mathematically searchable knowledge base.



Five Steps. Run Once.

The offline process that builds the vector store.

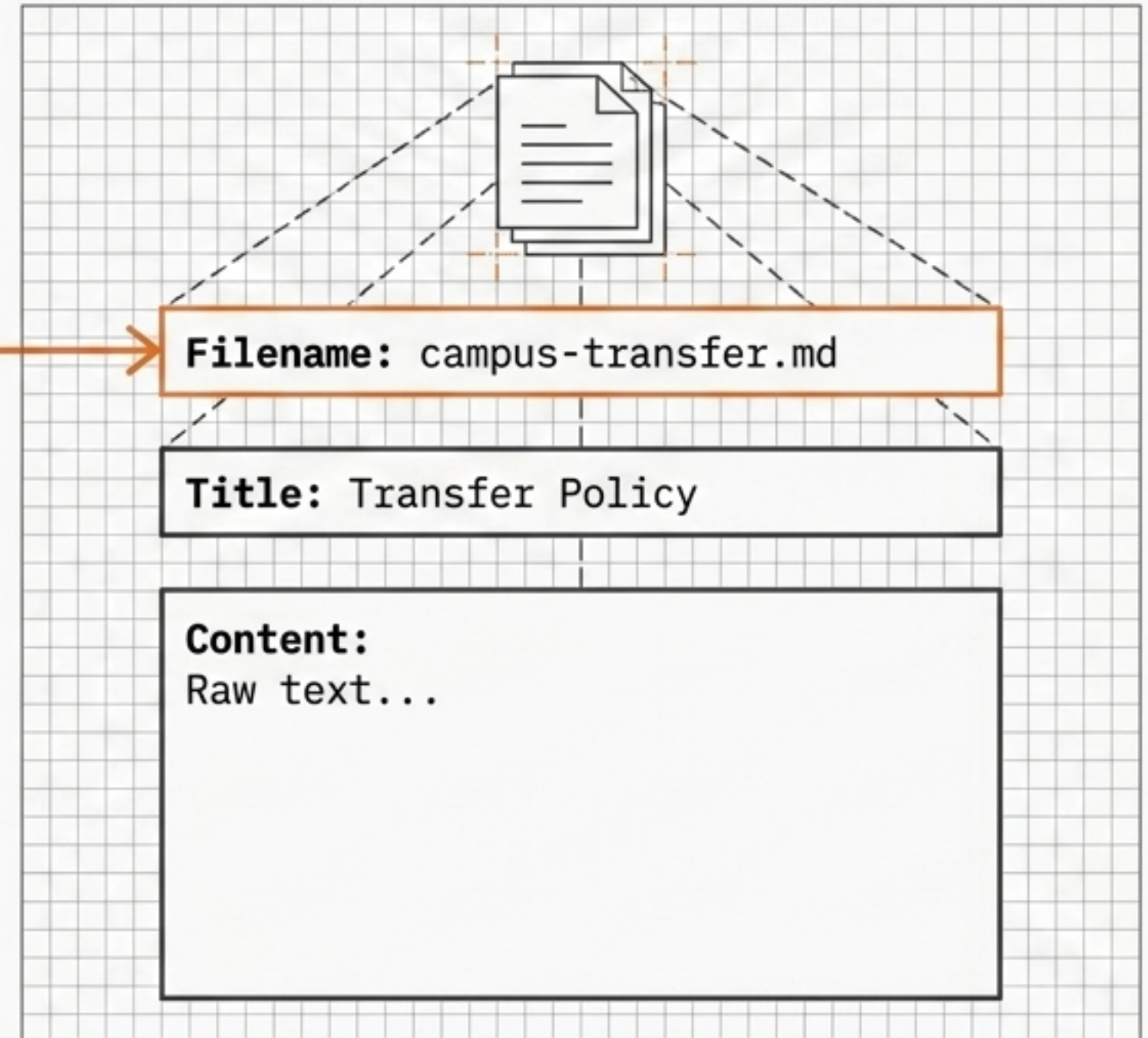


Extracting Content and Context

Reading raw text while preserving the source of truth.

1. Walk the directory for 91 markdown files.
2. Strip front-matter.
3. Store filename for exact source attribution.

Crucial link back
to the source.



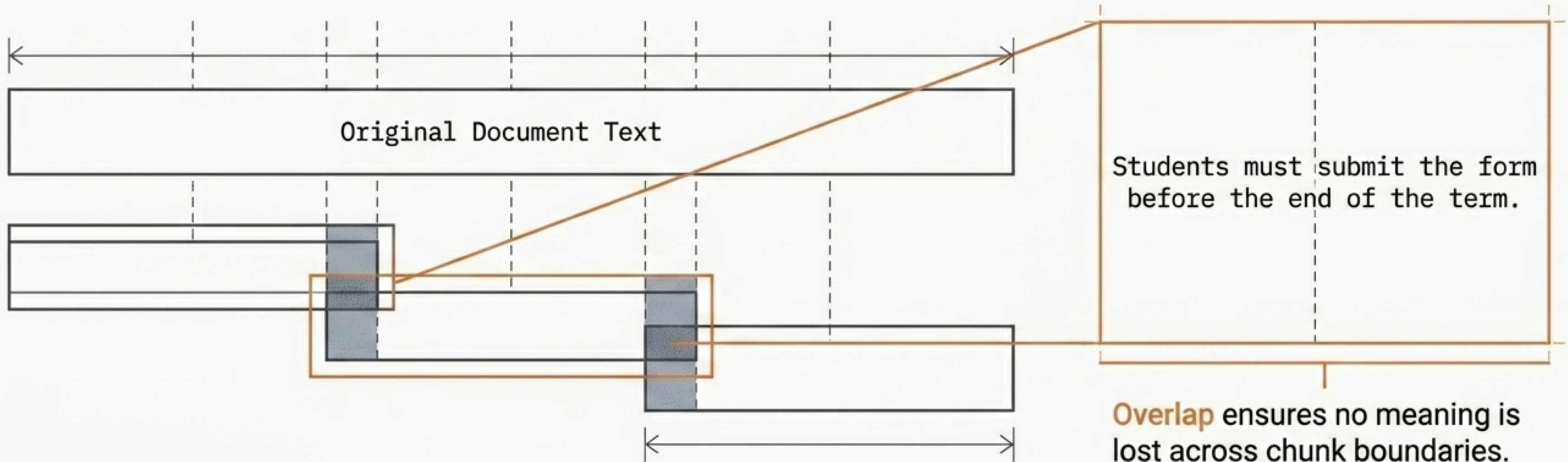
Beating the Context Limit

Splitting documents into precise, LLM-digestible pieces.

Strategy: 512 tokens per chunk

Buffer: 64 token overlap

Goal: Retain boundary context



Tokens ≠ Words

The fundamental reading unit of a language model.

1 Token $\approx$ 3 to 4 English characters.
Counted precisely using OpenAI's
tiktoken.

Hello

1 word = 1 token

Ho1

bert

on

1 word = 3 tokens

ingest

ion

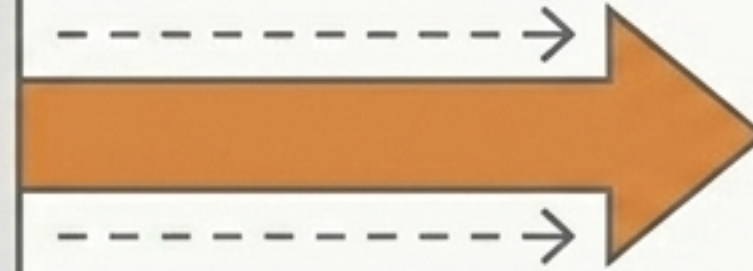
pipeline

2 words = 3 tokens

Converting Meaning into Math

Translating text chunks into a 768-dimensional space.

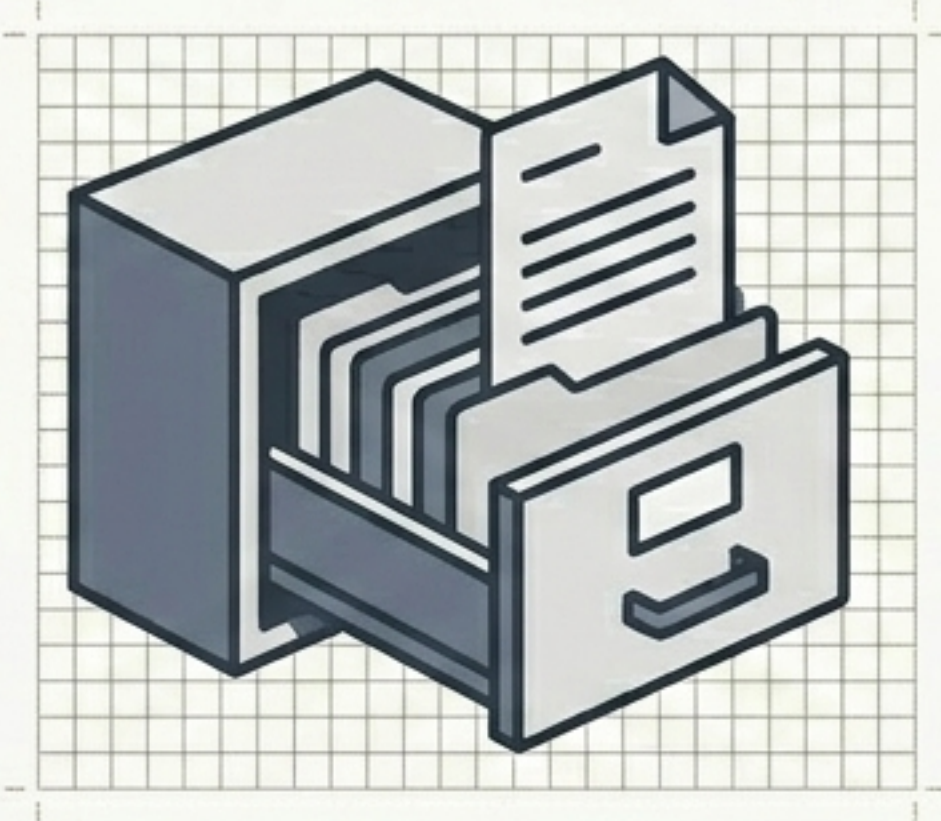
Campus transfer
policy requires
approval...


$$\begin{bmatrix} 0.045, \\ -0.112, \\ 0.884, \\ -0.331, \\ \dots 768 \text{ total} \end{bmatrix}$$

Model: Google gemini-embedding-001

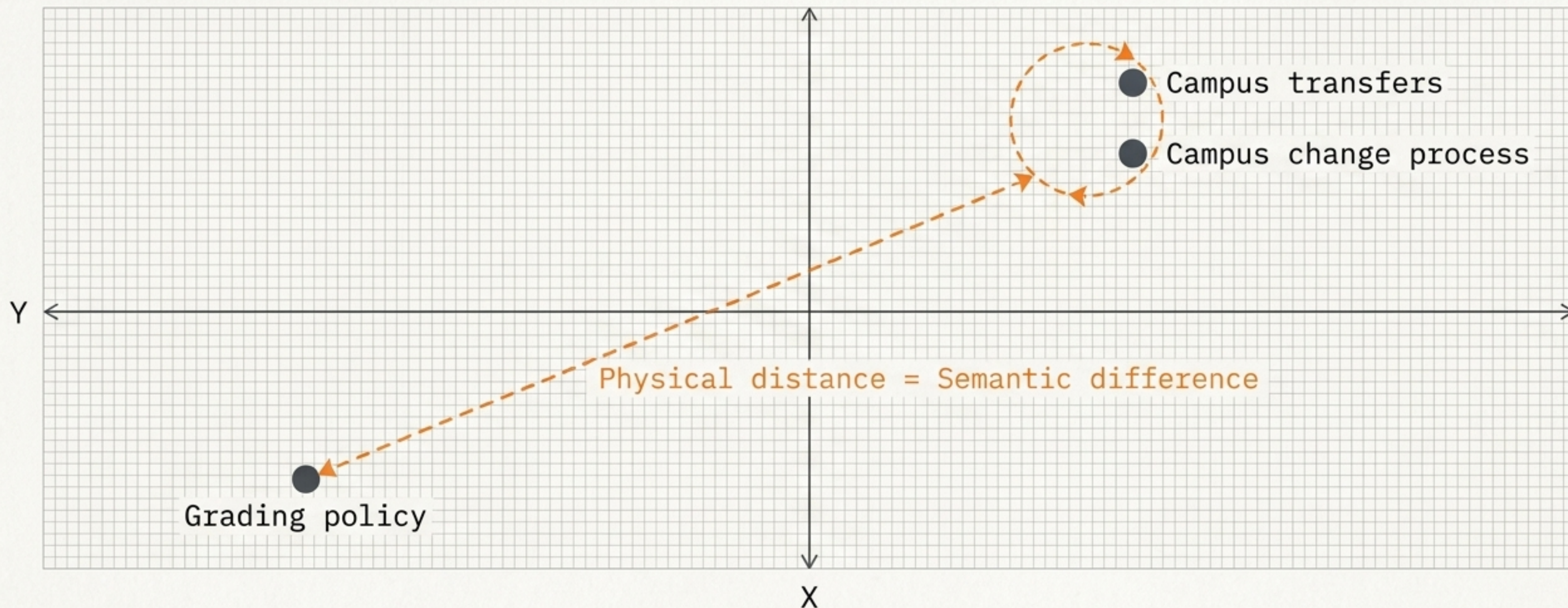
The Most Common RAG Mistake

Vectors must be optimized for their specific role in the system.
Using the wrong mode silently degrades retrieval quality.

RETRIEVAL_DOCUMENT	RETRIEVAL_QUERY
	
Used during the offline indexing phase. Optimizes the vector to BE FOUND.	Used live when a user asks a question. Optimizes the vector for FINDING.

Distance Equals Similarity

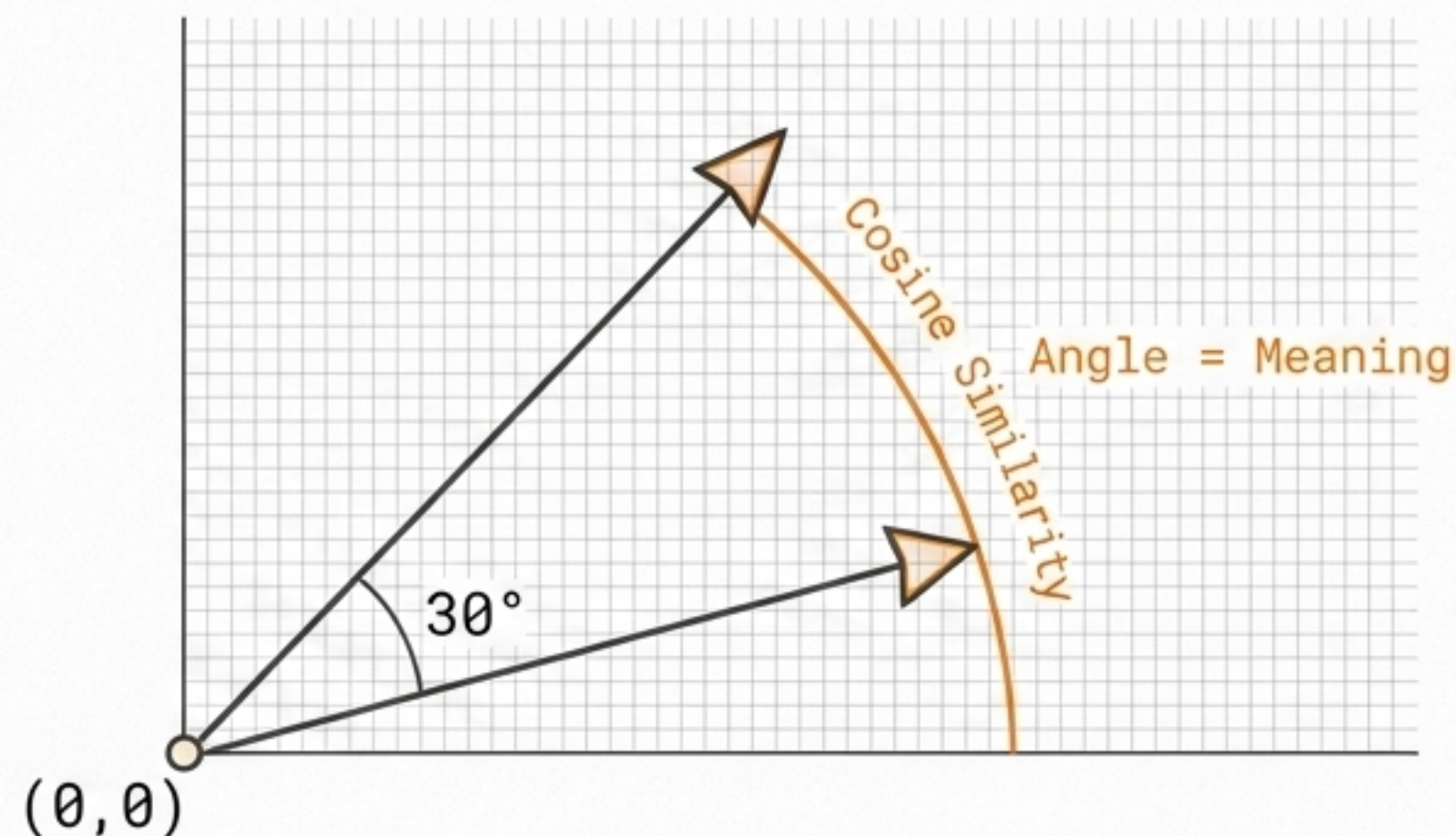
We don't read the 768 numbers; we measure the mathematical space between them.



High-Speed Semantic Search

Loading vectors into Facebook AI Similarity Search (FAISS).

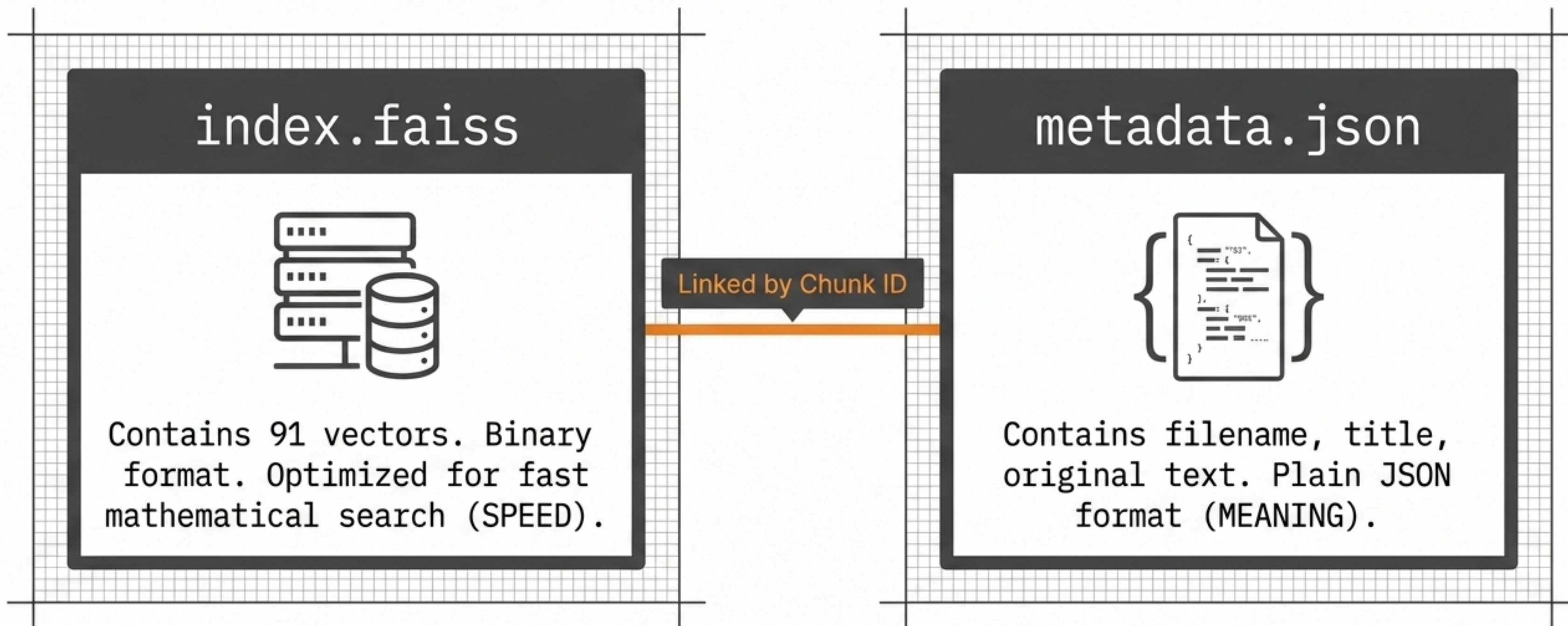
Index:	IndexFlatIP (No compression)
Math:	Vectors normalized to length 1
Engine:	Inner product perfectly calculates Cosine Similarity



Score 1.0 = Identical Meaning
Score 0.0 = Unrelated

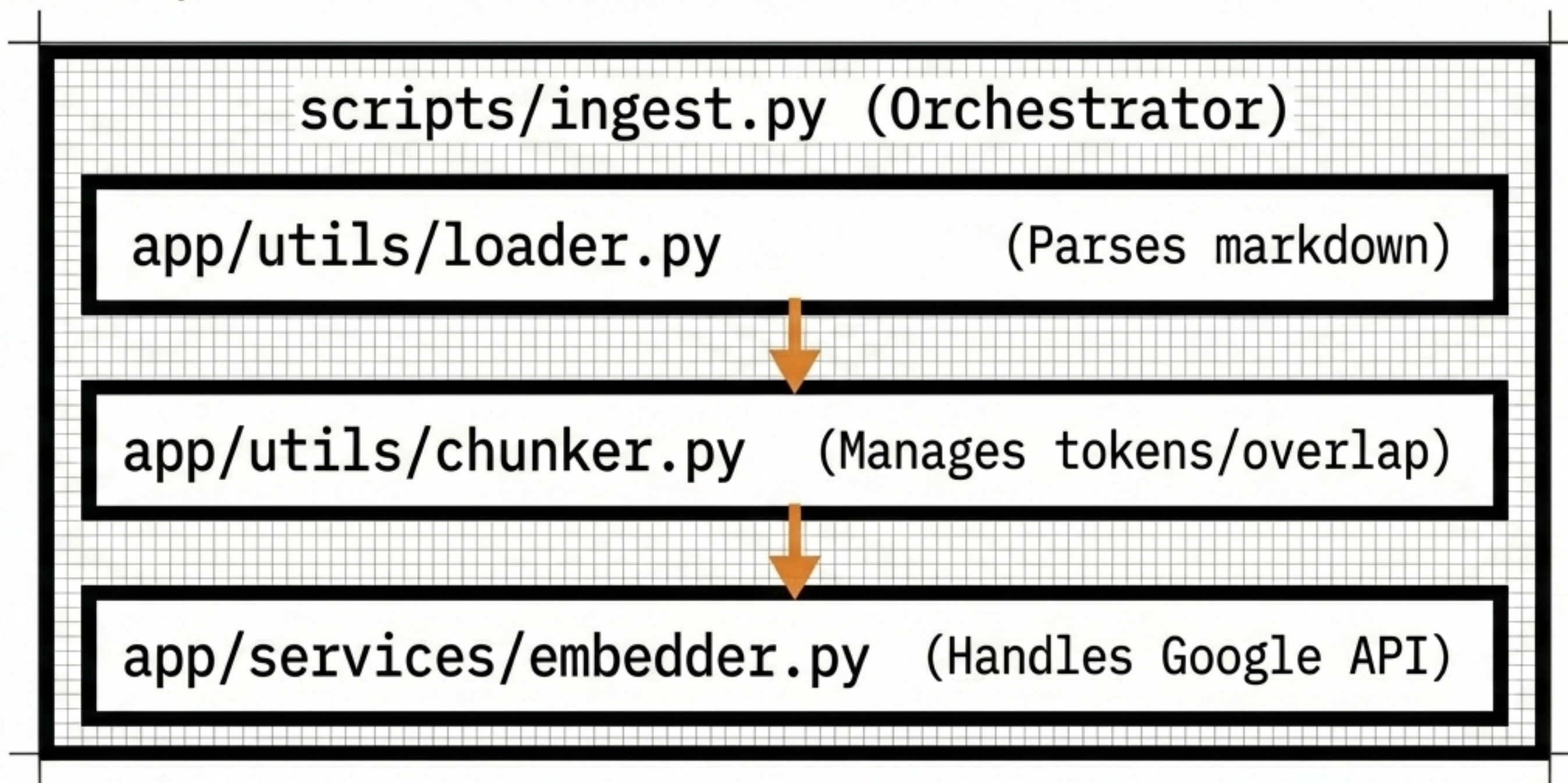
The Complete Knowledge Base

Search hits the binary index. Results map back to human-readable metadata.



Separation of Concerns

Every script has exactly one specific responsibility. Components are independently testable and replaceable.



Securing the Environment

Hardcoding API keys is an engineering anti-pattern.

Environment variables dictate models, paths, and chunk sizes outside the codebase.



The Foundation is Ready

91 documents successfully parsed, chunked, embedded, and indexed.

> Query: How can I transfer to another campus?

Searching FAISS index... OK

[MATCH FOUND] Source: campus-transfer.md | Similarity: 0.88

The Rules of Ingestion

Three principles to carry into every RAG project.



Offline Heavy Lifting

Run ingestion once. Ensure retrieval is instant because the hard mathematical work is already completed.



Chunking is Crucial

Size dictates precision of retrieval. Overlap dictates the retention of context at the boundaries.



Embeddings are Specific

Document vs. Query modes dictate search quality. Decisions made here compound throughout the entire system.